

Coding & Solution

Project: Rising Water – AI-Based Flood Prediction System

1. Solution Overview

The application uses a trained XGBoost machine learning model with a Flask web application to predict flood probability from 20 environmental parameters.

2. Development Process

1. Collect and preprocess the Kaggle flood dataset.
2. Split the dataset into training and testing sets.
3. Train the XGBoost model and save it using Joblib.
4. Develop the Flask backend.
5. Design the HTML/CSS frontend.
6. Integrate the model with the web application.
7. Test the application using sample inputs.

3. Modules

Module	Purpose
Dataset	Stores training data
Model	Predicts flood probability
Flask Backend	Handles requests
Frontend	Collects user inputs
Prediction Output	Displays probability and risk level

4. Technologies Used

- Python
- Flask
- Pandas
- NumPy
- Scikit-learn
- XGBoost
- HTML
- CSS
- Joblib

5. Conclusion

The coding approach follows a modular design that is easy to maintain and extend while providing accurate flood prediction.